

Cyber Security Mini Project : Test Password Strength

The Password Strength Checker is a Python-based program developed to analyze how strong a user's password is. The program asks the user to enter a password and then evaluates it based on predefined security rules.

The password is checked for minimum length, use of uppercase and lowercase letters, numbers, and special characters. The program also verifies whether the password avoids common patterns such as simple words or number sequences. Based on these checks, the program classifies the password as Weak, Medium, Strong, or Very Strong. If the password is weak, the program provides suggestions to improve password strength, helping users create more secure passwords.

This mini project helps users understand the importance of strong passwords and promotes better password practices for cybersecurity.

```
import re
import tkinter as tk

def check_strength():
    password = entry.get()
    score = 0
    suggestions = []

    if len(password) >= 8:
        score += 1
    else:
        suggestions.append("Use at least 8 characters")

    if re.search(r"[A-Z]", password):
        score += 1
    else:
        suggestions.append("Add uppercase letters (A-Z)")

    if re.search(r"[a-z]", password):
        score += 1
    else:
        suggestions.append("Add lowercase letters (a-z)")

    if re.search(r"[0-9]", password):
        score += 1
```

```

else:
    suggestions.append("Add numbers (0-9)")

if re.search(r"[@$!%*?&]", password):
    score += 1
else:
    suggestions.append("Add special symbols (@, $, !, etc.)")

if score <= 2:
strength = "Weak" color =
"red"
    elif score == 3:
        strength = "Medium"
        color = "orange"
    elif score == 4:
        strength = "Strong"
        color = "blue"
    else:
        strength = "Very Strong"
        color = "green"

result_label.config(text=f"Password Strength: {strength}", fg=color)

if strength in ["Weak", "Medium"]:
    suggestion_text = "Suggestions:\n"
    for s in suggestions:
        suggestion_text += "- " + s + "\n"
    suggestion_label.config(text=suggestion_text, fg="black")
else:
    suggestion_label.config(
        text="Good password! No improvement needed 🍌",
        fg="green"
    )

```

UI Window

```
window = tk.Tk()
window.title("Password Strength Checker")
window.geometry("450x350")

tk.Label(window, text="Enter Password", font=("Arial", 12)).pack(pady=10)

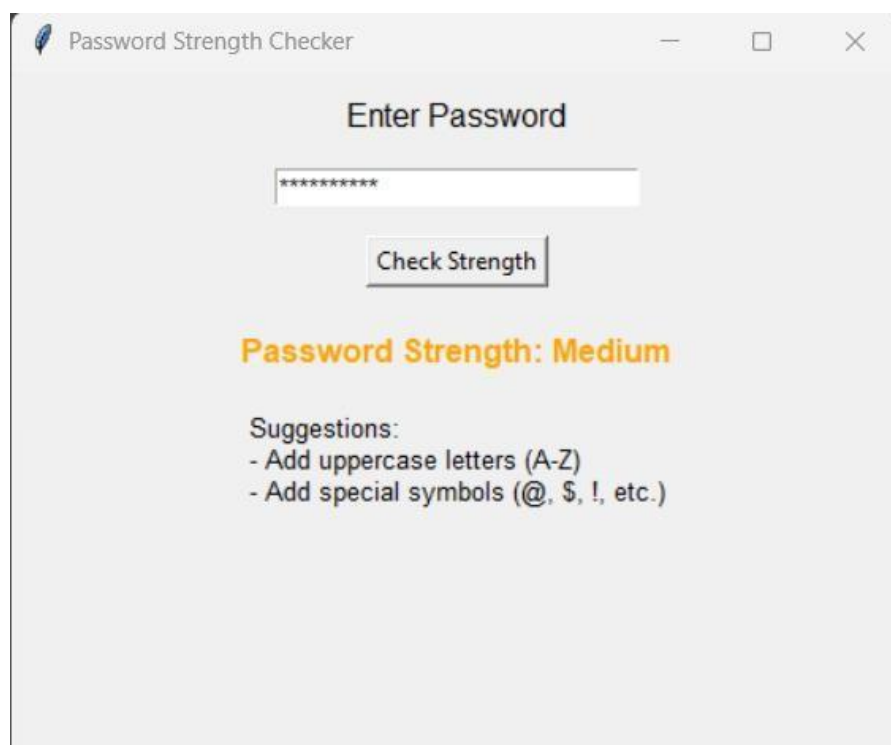
entry = tk.Entry(window, show="*", width=30)
entry.pack(pady=5)

tk.Button(window, text="Check Strength", command=check_strength).pack(pady=10)

result_label = tk.Label(window, text="", font=("Arial", 12, "bold"))
result_label.pack(pady=10)

suggestion_label = tk.Label(window, text="", justify="left", font=("Arial", 10))
suggestion_label.pack(pady=5)

window.mainloop()
```



Interface of a Password Strength Checker

